

Technology Stack

Date	5th July 2026
Team ID	RAHMA HUSSEIN JUMA
Project Name	CREDIT CARD APPROVAL PREDICTION
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS (Jinja2 templates via Flask)	Simple, lightweight forms are enough for data entry and results display — no complex client-side interactivity is needed, so a full framework like React wasn't necessary.

2	Backend / Server-Side	Python, Flask	Python is the same language used for the entire ML pipeline (pandas, scikit-learn, XGBoost), so the trained model can be loaded and used directly without converting between languages. Flask is lightweight and simple to set up for a small-scale prediction app.
3	Database / Data Storage	CSV files (application_record.csv, credit_record.csv) locally; no live database in the current version	The dataset is static and used for training rather than live transactional data, so flat files were sufficient. A relational database (matching the ERD's Users, Applicant_Details, Credit_History, Approval_Prediction tables) would be the natural next step if extended to a multi-user production system.
4	Cloud / Hosting / Deployment	IBM Watson Machine Learning (optional)	Allows the trained model to be hosted on the cloud instead of only running locally, enabling scalable, real-time predictions accessible from anywhere, as outlined in the original project brief.

5	Version Control & CI/CD	GitHub	Used to store, version, and submit the project; provides a single accessible link for the professor/mentor to review the full codebase and documentation.
6	Third-Party APIs / Other Tools	IBM Watson Machine Learning API	Used specifically to deploy and serve the trained model as a hosted scoring endpoint, instead of (or alongside) the local pickled model in Flask.